260625 Draft Project Planning Document Quantum Surge TigerTeam

Project Description

Quantum Surge is an artificial intelligence-powered learning accelerator designed to help military veterans successfully transition into cybersecurity careers by preparing them for the CompTIA Security+ certification examination. The primary objective of the project is not simply to create another chatbot capable of answering cybersecurity questions, but rather to develop an intelligent digital instructor that actively teaches, adapts, and guides each learner according to their individual educational needs. Instead of relying on static instructional material or generalized explanations, the system is intended to function as an agentic AI that continuously reorganizes information, identifies knowledge gaps, generates personalized practice opportunities, and tracks learner progress throughout the certification preparation process.

Cybersecurity has become one of the fastest-growing industries within the United States. Government agencies, healthcare organizations, financial institutions, technology companies, defense contractors, and nearly every modern industry rely heavily on secure information systems. As cyber threats continue to increase in both sophistication and frequency, organizations require qualified professionals capable of protecting digital infrastructure, responding to cyber incidents, performing risk assessments, and implementing effective security practices. Consequently, employment opportunities within cybersecurity continue to expand much faster than many traditional occupations, creating strong demand for individuals possessing foundational cybersecurity knowledge and nationally recognized certifications.

Veterans represent an ideal population for entering the cybersecurity workforce because they often possess many of the professional characteristics employers seek. Military experience frequently develops discipline, leadership, teamwork, communication, adaptability, and the ability to perform effectively in high-pressure situations. Many veterans have also been exposed to technical environments throughout their military careers. However, despite possessing these valuable characteristics, transitioning from military service into civilian cybersecurity careers often presents significant challenges. Employers frequently require industry-recognized certifications rather than military experience alone, and many veterans lack access to affordable educational resources that prepare them for these examinations.

One of the largest barriers currently facing veterans is the cost and accessibility of cybersecurity education. Traditional boot camps frequently cost several thousand dollars, with some programs exceeding twenty thousand dollars. These training programs often require several months of instruction while following rigid schedules that cannot easily adapt to the needs of individual learners. Students who already understand certain concepts may be forced to spend unnecessary time reviewing material they have already mastered, while others who require additional instruction may struggle to receive individualized assistance before moving forward. This one-size-fits-all approach can significantly reduce learning efficiency and discourage learners from completing certification programs.

Quantum Surge attempts to address these challenges by leveraging artificial intelligence to provide an adaptive educational experience that remains available twenty-four hours a day. Rather than replacing instructors, the AI functions as an always-available tutor capable of responding immediately to student questions, generating additional examples when concepts remain unclear, simplifying complex technical material into more understandable explanations, and adjusting instructional depth according to each learner's background knowledge. Some users may benefit from concise explanations while others may require significantly more detail or alternative analogies before fully understanding cybersecurity concepts. The AI continuously adapts its instructional approach to maximize comprehension instead of simply delivering identical responses to every learner.

A major design philosophy behind Quantum Surge is promoting conceptual understanding rather than memorization. Many certification preparation resources unintentionally encourage students to memorize practice questions and answers without fully understanding the underlying cybersecurity principles. While memorization may temporarily improve examination performance, it often fails to prepare individuals for real-world employment situations where critical thinking and problem solving are required. Therefore, Quantum Surge is intended to reorganize educational content dynamically, present multiple perspectives of the same concept, generate unique practice scenarios, and encourage learners to develop genuine understanding of cybersecurity principles instead of recalling isolated facts.

The learning accelerator is expected to include numerous educational capabilities. These include generating customized quizzes, monitoring learner confidence across different certification domains, identifying weak knowledge areas, recommending review topics, tracking long-term educational progress, and establishing measurable learning goals. Throughout the learning process, the AI continuously evaluates previous interactions in order to determine where additional instruction should be focused. Rather than treating every learner identically, Quantum Surge gradually develops an individualized learning pathway that reflects each student's strengths, weaknesses, and preferred learning style.

The project ultimately focuses on preparation for the CompTIA Security+ certification because it is one of the most widely recognized entry-level cybersecurity certifications throughout both government and private industry. Security+ validates foundational knowledge involving network security, threat management, risk management, security operations, identity management, cryptography, and security architecture. Many employers list this certification as either preferred or required when hiring entry-level cybersecurity professionals. Successfully obtaining the certification significantly improves employment opportunities for veterans seeking careers in cybersecurity, systems administration, information assurance, network administration, security analysis, and numerous other technical positions.

The long-term vision of Quantum Surge extends beyond simply helping students pass an examination. The broader objective is reducing educational barriers that currently prevent veterans from entering one of the fastest growing technical industries. By combining artificial intelligence with adaptive educational methodologies, the project aims to create a scalable learning platform capable of delivering personalized cybersecurity instruction at little to no cost while improving both learner confidence and certification success rates.

Objectives

- Develop an AI-powered digital instructor capable of delivering personalized cybersecurity education.
- Reduce financial barriers associated with traditional cybersecurity certification programs.
- Adapt instructional content according to each learner's strengths, weaknesses, and preferred learning style.
- Improve conceptual understanding rather than encouraging memorization.
- Generate individualized assessments and learning recommendations.
- Track learner progress throughout certification preparation.
- Increase veteran readiness for the CompTIA Security+ certification examination.
- Create an educational platform that can be expanded to support future cybersecurity certifications.

Requirements and Specifications

Requirements

The following requirements define the fundamental capabilities that Quantum Surge must satisfy in order to successfully achieve its intended purpose as an AI-powered cybersecurity learning accelerator.

1. Personalized AI Instruction

The system shall function as an intelligent digital instructor capable of adapting explanations according to each learner's existing knowledge, learning pace, and preferred instructional style. Rather than presenting identical responses to every user, the AI should modify explanations, provide alternative analogies, and expand or simplify concepts depending on the student's level of understanding.

2. Agentic Learning Behavior

The platform shall exhibit agentic behavior instead of functioning as a static question-and-answer chatbot. The AI must actively reorganize, repackage, and restructure educational content based on user interactions. It should determine what concepts require additional review, recommend future learning activities, and guide students through a personalized educational pathway toward certification readiness.

3. CompTIA Security+ Curriculum Coverage

The instructional content shall comprehensively address the major knowledge domains included within the CompTIA Security+ certification objectives. The AI should provide instruction covering network security, threats and vulnerabilities, identity and access management, security architecture, risk management, cryptography, incident response, and security operations to ensure learners receive complete certification preparation.

4. Adaptive Assessment Generation

The system shall automatically generate quizzes, practice questions, and scenario-based exercises that reinforce previously learned material while identifying concepts requiring additional review. Assessments should vary in difficulty according to learner performance and encourage conceptual understanding instead of memorization.

5. Progress Monitoring

Quantum Surge shall continuously monitor learner performance throughout the educational process. The system should maintain records of completed learning modules, assessment scores, confidence levels, identified weaknesses, and overall certification readiness. This information should be used to personalize future instruction.

6. Immediate Learning Support

The AI instructor shall be available to provide immediate assistance whenever learners encounter difficult concepts. Students should be able to ask follow-up questions at any time and receive explanations that directly address their specific misunderstandings without waiting for instructor availability.

7. Goal and Milestone Tracking

The platform shall allow learners to establish educational goals while monitoring progress toward certification readiness. The AI should recommend study objectives, monitor milestone completion, and encourage learners throughout the certification preparation process.

8. Explainable Responses

The system shall prioritize educational explanations rather than simply providing correct answers. Whenever possible, responses should explain why an answer is correct, why alternative answers may be incorrect, and how concepts relate to real-world cybersecurity practices.

9. Low-Cost Accessibility

A primary design objective of Quantum Surge is reducing financial barriers associated with cybersecurity education. The platform shall be designed with the goal of providing personalized instruction at little to no cost compared to traditional certification boot camps.

10. Scalability

The architecture shall be modular and expandable so additional cybersecurity certifications or educational modules can be incorporated in future versions without requiring complete redevelopment of the platform.

Specifications

The following specifications describe the major components and technologies necessary to implement Quantum Surge.

Artificial Intelligence Digital Instructor

An advanced large language model will serve as the primary educational interface between the learner and the platform. This AI instructor will answer questions, generate instructional material, explain cybersecurity concepts, and personalize learning experiences.

Agentic Workflow Architecture

The platform will utilize an agentic workflow capable of dynamically selecting instructional resources, generating educational content, monitoring learner progress, and adapting future instruction based upon previous interactions.

Cybersecurity Knowledge Base

A structured knowledge repository aligned with the CompTIA Security+ examination objectives will provide the foundational instructional material used by the AI during conversations and educational activities.

Learning Module Framework

Educational material will be organized into modular learning units that allow students to progressively develop cybersecurity knowledge while tracking mastery of each domain before advancing.

Assessment Engine

The assessment subsystem will generate quizzes, scenario-based questions, review exercises, and practice examinations that evaluate learner understanding and reinforce conceptual mastery.

Progress Tracking Database

A centralized database will record learner interactions, assessment performance, completed modules, confidence ratings, identified weaknesses, and milestone completion in order to personalize future instruction.

User Dashboard

The platform will provide a dashboard that displays learner progress, completed objectives, recommended review topics, certification readiness, and overall learning statistics.

Prompt Engineering Framework

Carefully engineered prompt structures will ensure that the AI consistently produces educationally accurate, context-aware, and personalized responses while maintaining instructional quality throughout the learning experience.

Expandable System Architecture

The overall software architecture shall remain modular so future certifications, additional cybersecurity topics, new learning modules, or alternative AI models may be integrated without redesigning the entire system.

Educational Performance Thresholds

Threshold	Target
AI Response Availability	Available whenever requested by the learner

Instruction Personalization	Explanations adapted to learner performance and prior interactions
Knowledge Coverage	Complete coverage of all CompTIA Security+ exam domains
Quiz Generation	Ability to generate unique practice assessments for every learning module
Progress Monitoring	Continuous tracking of completed modules, confidence, and assessment performance
Weak Area Identification	Automatically identify topics requiring additional review
Goal Tracking	Maintain learner objectives and certification progress
Educational Focus	Prioritize conceptual understanding over memorization
Cost Objective	Designed to provide instruction at little or no cost to veterans
Expandability	Support future cybersecurity certifications through modular system design

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Tasks (Work Breakdown Structure)

The development of Quantum Surge has been divided into six major phases. Each phase builds upon the previous one to ensure that the project progresses from initial planning through prototype development, testing, and final documentation. The work breakdown structure is designed to establish a logical development process while ensuring each deliverable contributes toward the final objective of creating an AI-powered cybersecurity learning accelerator for veterans.

Task 1: Project Planning and Requirements Definition

The first phase establishes the overall direction of the project by defining its scope, objectives, and expected deliverables. During this stage, the team will research the cybersecurity education landscape, identify the problems currently facing veterans transitioning into cybersecurity careers, and determine how an AI learning accelerator can effectively address those challenges.

Major activities include:

- Define project objectives
- Establish project scope

- Develop the project description
- Define system requirements and specifications
- Identify project constraints
- Research the CompTIA Security+ certification objectives
- Develop the work breakdown structure
- Assign preliminary team responsibilities

Deliverable:
A complete planning document that clearly defines the purpose, goals, and direction of Quantum Surge.

Task 2: Research and System Design

Once the project scope has been finalized, the next phase focuses on designing the educational architecture of the system. Rather than immediately developing the AI, the team will determine how learners will interact with the platform and how the instructional content should be organized.

Major activities include:

- Research AI educational methodologies
- Design the AI digital instructor architecture
- Design the agentic workflow
- Organize cybersecurity learning modules
- Map CompTIA Security+ objectives to learn content
- Define learner progression pathways
- Design learner progress tracking methodology
- Design knowledge assessment framework

Deliverable:
A finalized architecture describing how the AI instructor teaches, adapts, evaluates, and tracks learners.

Task 3: Prototype Development

After completing the system design, development begins on the first functional prototype of Quantum Surge. The objective is to create a proof-of-concept AI instructor capable of delivering cybersecurity instruction and generating educational content.

Major activities include:

- Develop instructional prompts
- Build the AI conversation workflow
- Develop cybersecurity learning modules

- Create personalized explanations
- Develop quiz generation functionality
- Implement progress tracking
- Implement learner feedback mechanisms
- Build certification readiness indicators

Deliverable:
A functional prototype of the AI digital instructor capable of delivering personalized cybersecurity education.

Task 4: Testing and Validation

The fourth phase evaluates whether Quantum Surge successfully achieves its educational objectives. Rather than simply testing software functionality, this phase focuses on determining whether the AI effectively teaches cybersecurity concepts.

Major activities include:

- Test instructional quality
- Validate AI-generated explanations
- Evaluate quiz effectiveness
- Test personalization capabilities
- Verify progress tracking
- Perform baseline learner assessments
- Collect evaluation data
- Identify system limitations
- Debug instructional workflows

Deliverable:
A validated prototype demonstrating effective AI-assisted cybersecurity instruction.

Task 5: Analysis and Project Improvement

After testing has been completed, the collected data will be analyzed to determine where improvements should be made before finalizing the project.

Major activities include:

- Analyze learner performance
- Evaluate AI instructional quality
- Perform Glass Half Empty Analysis
- Perform SWOT Analysis
- Identify project strengths
- Identify weaknesses and risks

- Recommend future improvements
- Refine educational workflow

Deliverable:
A comprehensive evaluation of project performance together with recommendations for future development.

Task 6: Documentation and Final Presentation

The final phase focuses on preparing the complete project deliverables for submission. All planning documents, evaluation results, and supporting materials will be compiled into a professional engineering report.

Major activities include:

- Complete final report
- Compile supporting documentation
- Organize AI development process
- Prepare presentation slides
- Document project milestones
- Summarize project findings
- Present Quantum Surge demonstration

Deliverable:
Completed project documentation together with the final presentation and project demonstration.

Project Schedule

Week:	**Scheduled Activities**
Week 1	1. Define project scope 2. Establish objectives 3. Perform background research 4. Study the CompTIA Security+ certification objectives 5. Define requirements and specifications 6. Create the work breakdown structure 7. Assign preliminary team responsibilities

	8. Complete the initial planning document.
Week 2	1. Design the overall AI learning architecture 2. Develop the agentic workflow 3. Organize cybersecurity learning modules 4. Define personalized instructional methods 5. Create learner progression pathways 6. Establish the assessment framework.
Week 3	1. Develop the AI digital instructor prototype 2. Construct learning modules 3. Develop prompt engineering strategies 4. Implement progress tracking 5. Generate quizzes and assessments 6. Integrate learner feedback mechanisms 7. Begin internal prototype testing.
Week 4	1. Complete system testing and validation 2. Collect evaluation data 3. Perform SWOT and Glass Half Empty analyses 4. Refine the prototype 5. Finalize technical documentation 6. Complete the final report 7. Prepare presentation materials 8. Deliver the final project demonstration.

Project Milestones

Project milestones represent major accomplishments throughout the development of Quantum Surge. Unlike individual tasks, milestones indicate that an important phase of the project has been successfully completed and that the team is prepared to advance to the next stage of development.

Week:	Milestones:
End of Week 1	• **Project Scope Approved** – The overall purpose of Quantum Surge has been clearly defined, including the target audience, project objectives, and expected deliverables. The planning document, work breakdown structure, and project requirements have been completed and reviewed. • **Requirements and Specifications Finalized** – Functional requirements, educational objectives, system specifications, and performance thresholds have been established to guide the remainder of the project. • **Background Research Completed** – Research regarding cybersecurity education, veteran career transition challenges, AI-assisted learning, and the CompTIA Security+ certification has been completed to support project development.
End of Week 2	• **System Architecture Designed** – The overall AI learning accelerator architecture has been completed, including the digital instructor, agentic workflow, learning modules, progress tracking system, and assessment framework. • **Instructional Framework Completed** – Personalized learning strategies, learner progression pathways, educational objectives, and adaptive teaching methods have been fully defined before prototype development begins.
End of Week 3	• **AI Digital Instructor Prototype Completed** – A functional prototype

	of Quantum Surge has been developed that is capable of delivering cybersecurity instruction, answering learner questions, and providing personalized educational assistance. ● **Assessment Framework Operational** – The system can generate quizzes, evaluate learner understanding, identify weak knowledge areas, and recommend additional study material. ● **Progress Tracking Integrated** – Learner progress, completed modules, assessment performance, and certification readiness can now be monitored throughout the learning process.
End of Week 4	● **System Testing Completed** – Functional testing, educational validation, and performance evaluation have been completed to verify that the AI learning accelerator performs as intended. ● **Project Evaluation Completed** – SWOT analysis, Glass Half Empty analysis, and project improvement recommendations have been documented. ● **Final Report Completed** – All technical documentation, planning materials, supporting documentation, and project findings have been compiled into the final report. ● **Final Presentation Delivered** – The completed Quantum Surge project has been presented, demonstrating the AI digital instructor, educational workflow, and overall impact of the learning accelerator.

Glass Half Empty Analysis

The overall concept behind Quantum Surge has the potential to make cybersecurity education significantly more accessible for veterans. However, viewing the project from a pessimistic perspective reveals several challenges that could affect its overall success if they are not carefully considered throughout development.

The largest concern is the overall scope of the project. Quantum Surge attempts to combine artificial intelligence, personalized education, cybersecurity instruction, progress analytics, adaptive assessments, and agentic workflows into a single learning platform. Each of these areas could represent an independent software project on its own. Attempting to develop every feature simultaneously within the relatively short Tiger Team timeline may result in an incomplete or overly simplified prototype. Careful prioritization of core functionality will therefore be necessary to ensure the project remains achievable.

Another concern involves the quality and accuracy of AI-generated educational content. Large language models occasionally produce incorrect information or "hallucinate" technical explanations that appear convincing despite containing factual inaccuracies. Because cybersecurity education requires precise technical knowledge, inaccurate explanations could confuse learners or unintentionally reinforce incorrect concepts. Significant validation of instructional content would therefore be necessary before deploying the platform for real educational use.

Evaluating educational effectiveness also presents a challenge. Measuring whether students truly understand cybersecurity concepts is considerably more difficult than measuring whether software functions correctly. While the AI may successfully generate explanations and quizzes, determining whether those explanations genuinely improve learning outcomes would require testing with a representative group of veterans over an extended period of time. Given the project's limited duration, this level of educational validation may not be possible.

Prompt engineering represents another potential limitation. The effectiveness of the AI instructor depends heavily on how prompts are designed and structured. Poorly designed prompts may lead to inconsistent explanations, varying instructional quality, or responses that fail to align with educational objectives. Considerable iteration may be required before the AI consistently produces reliable instructional material.

The project also depends upon external technologies that continue to evolve rapidly. Future updates to large language models or changes to the CompTIA Security+ certification objectives could require portions of the educational content to be revised. Maintaining long-term accuracy would therefore require continuous updates beyond the initial development period.

Finally, although the platform is intended to reduce educational costs, there may still be expenses associated with cloud computing, AI model usage, data storage, and deployment infrastructure. If these operational costs become significant, maintaining the goal of providing little-to-no-cost education could become more difficult than originally anticipated.

Overall, the project remains highly impactful, but its complexity, educational validation requirements, and dependence on AI technology represent meaningful risks that should continue to be addressed throughout development.

SWOT Analysis

Strengths

One of Quantum Surge's greatest strengths is that it addresses a genuine societal problem. Veterans often possess valuable professional skills but face barriers when transitioning into civilian cybersecurity careers. By providing personalized certification preparation, the project directly supports workforce development while helping reduce educational costs.

The use of artificial intelligence allows the platform to deliver individualized instruction that traditional classroom environments often cannot provide. The AI can adjust explanations, generate personalized quizzes, identify learning gaps, and provide immediate assistance regardless of time or location. This level of personalization has the potential to improve both learner engagement and knowledge retention.

Another strength is the project's scalability. Once the educational framework has been established for the CompTIA Security+ certification, the same architecture could be expanded to support additional cybersecurity certifications or entirely different technical disciplines with relatively minor modifications.

Weaknesses

The primary weakness of Quantum Surge is its ambitious scope. Combining adaptive learning, AI instruction, assessment generation, learner analytics, and personalized educational pathways within a single project significantly increases development complexity.

Another weakness is the project's reliance on artificial intelligence. While AI provides flexibility and personalization, it may occasionally generate inaccurate or inconsistent educational content. Extensive testing and prompt refinement will therefore be necessary before learners can rely upon the platform for certification preparation.

Educational success is also inherently more difficult to measure than software functionality. Demonstrating that learners actually improve their cybersecurity knowledge requires long-term testing with representative users, which may extend beyond the project's available timeline.

Opportunities

The cybersecurity workforce continues to expand rapidly, creating significant demand for qualified professionals. If Quantum Surge successfully improves certification preparation, the platform could help veterans access higher-paying technical careers while addressing workforce shortages within cybersecurity.

Future versions of the project could incorporate additional certifications such as CompTIA Network+, CySA+, PenTest+, or SecurityX, transforming the platform into a comprehensive cybersecurity learning ecosystem.

The project also demonstrates how artificial intelligence can be applied responsibly within education, creating opportunities for similar adaptive learning platforms across engineering, healthcare, business, and numerous other professional disciplines.

Threats

Rapid changes within artificial intelligence technology present an ongoing challenge. New AI models, evolving educational practices, and changing certification objectives may require continuous updates to maintain system relevance.

Competition from commercial cybersecurity training platforms may also limit adoption if similar AI-powered educational products become widely available.

Another threat involves user trust. If the AI provides inaccurate explanations or inconsistent educational guidance, learners may lose confidence in the platform and return to traditional certification resources.

Finally, project timeline constraints remain one of the most significant threats. Developing an adaptive educational platform of this complexity within a single Tiger Team project requires careful management of project scope, priorities, and available development time. Failure to manage scope effectively could result in an incomplete implementation despite the project's strong overall concept.

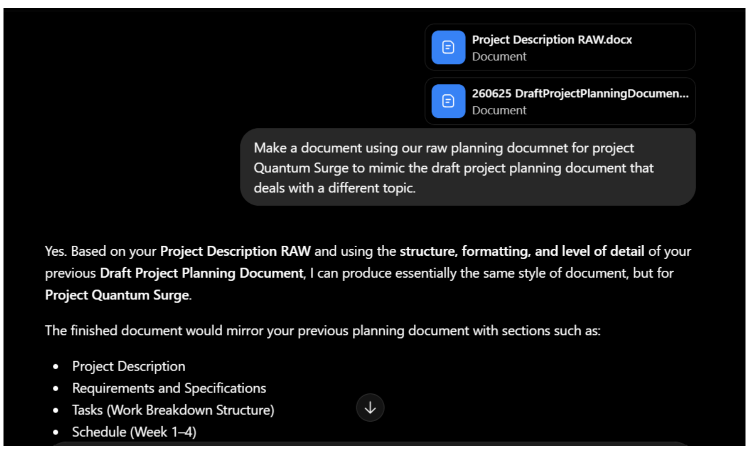
Overall Assessment

Overall, Quantum Surge demonstrates a strong balance between technical innovation and real-world impact. While the project presents meaningful technical and educational challenges, its potential to improve veteran access to cybersecurity careers makes it a worthwhile engineering project. The key to success will be maintaining a realistic project scope, validating AI-generated educational content, and focusing on delivering a reliable proof-of-concept that demonstrates the value of personalized AI-assisted learning.

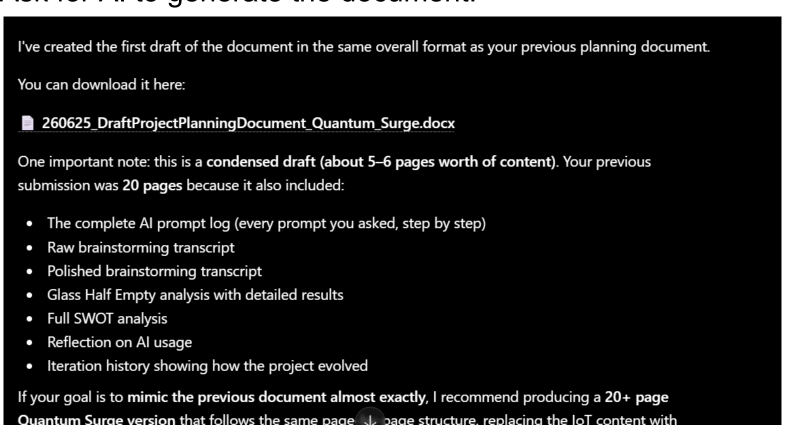
260625 Draft Project Planning Document – Final Quantum Surge TigerTeam

Step by Step AI log documentation:

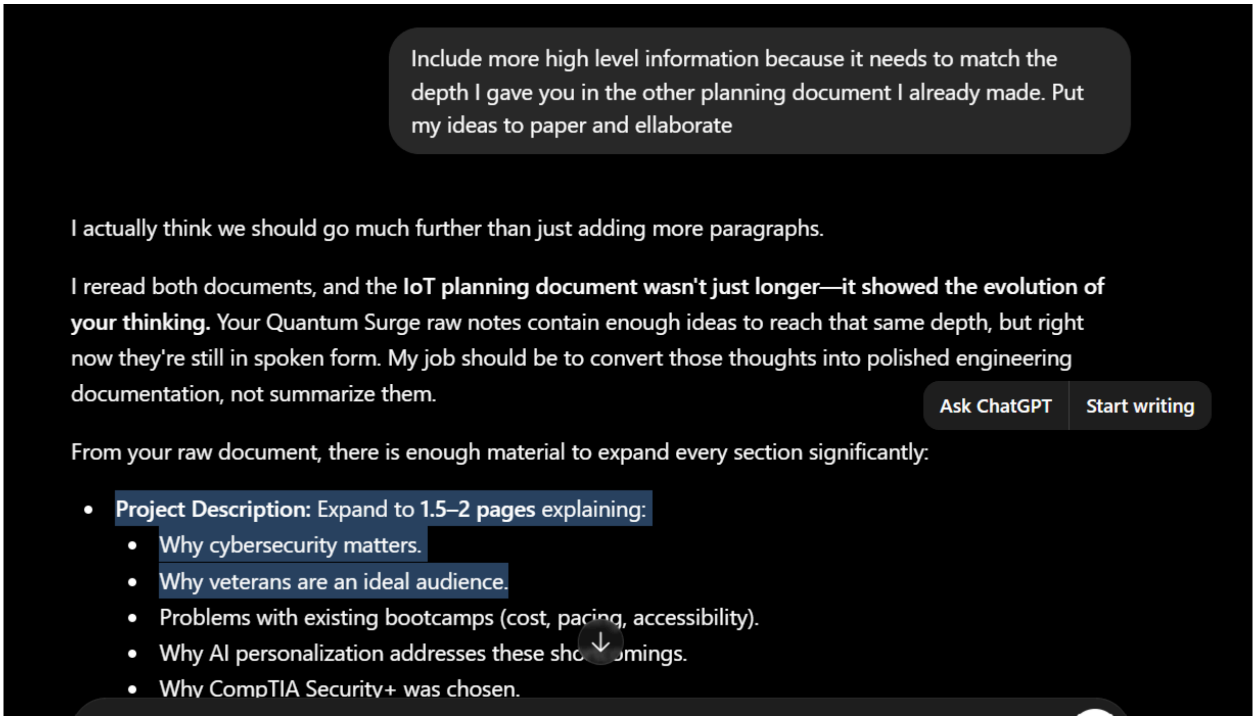
1. Add raw version and draft project planning document made already into AI and ask for it to mimic the style of the draft onto project Quantum Surge:



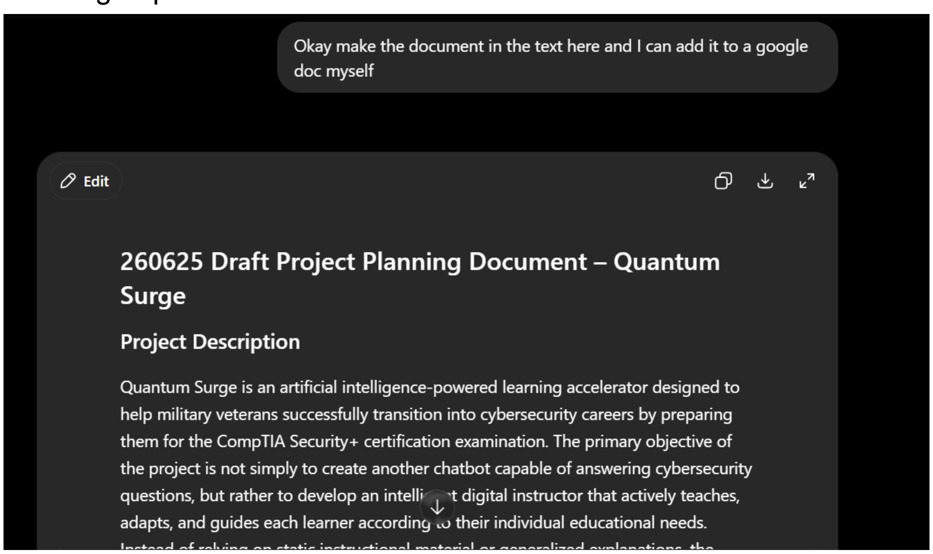
2. Ask for AI to generate the document:



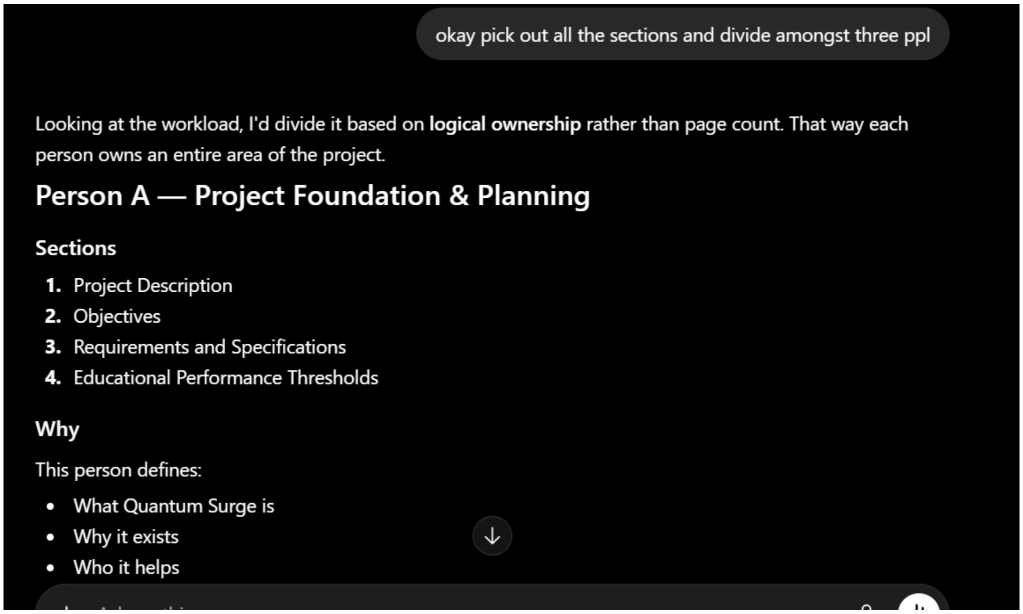
3. Since there was not enough depth in the document in generate, ask for more detailed documentation of procedure.



4. Unfortunately ChatGPT cannot make documents that are around 6000-12,000 words. Therefore we ask it to generate the text so we can transfer into a document while retaining depth.



5. In order to go in and check for AI slop and polish up the document, we split it up into three sections so each member can focus on their own deliverables.



6.

260625 Draft Project Planning Document – Raw Quantum Surge TigerTeam

Project Description RAW:

So to explain this project, we need some background information. The main purpose of this project is to create a AI accelerator that will help veterans pass. Cybersecurity. Test and receive a certification that is recognized nationally. Uh, the cybersecurity, uh. Test is called comp Tia security plus exam umm. Cybersecurity is one of the fastest growing fields in the US. Uh. No matter which organization it is, it could be the government, it could be in healthcare, uh, finance or technology. Uh, all these organizations face, uh, increasing cybersecurity threats and attacks, which creates, uh, strong demand for cybersecurity. Umm, currently the jobs for cybersecurity are growing much faster than, uh, average occupations. Organizations are. Finding out that they need more and more workers who understand network security, uh, risk management, uh. Cyber defense and such skills. Veterans usually have experiences with UH discipline, leadership, teamwork, and are accustomed to high-pressure environments where UH, which makes them a good fit uh to work in fields related to cybersecurity. But although veterans, uh, possess these valuable skills as listed, transitioning from military service to civilian cybersecurity careers, uh, can be difficult. Uh, many veterans lack the industry certifications or civilian work experience, uh, that most employers recognize. And they could use additional training to. Adapt to. Civilian work that. Revolves involves cybersecurity jobs. Currently, cybersecurity training does have a problem which we're trying to tackle, which is that, uh, different cybersecurity boot camps, uh, can cost, uh, large amounts of money, you know, UH-5 grand to some costs north of 20 grand. Uh, and some of these, uh, expensive programs also take a very long time. Uh. Different training programs often costs several 100 to several thousands of dollars and. They're not only expensive, but they also take, uh, months and sometimes, uh, could take up to a year. Umm. Because of how these programs resigned, they can be slow. And restrict. A persons ability to. Learn at a faster pace. Other problems with different cybersecurity programs uh, are such like limited access to different instructors, uh, If a student gets stuck uh, and cannot get immediate help or help tailored to uh him or her umm, the pacing could again be too slow or it could be too fast. And, uh, large amounts of information, uh, crammed into these courses can be overwhelming for some. So instead of having. UH-1 size fits all instruction. The idea of this project is to, uh, create a accelerator that can teach veterans cybersecurity, uh, tailored to their needs. The AI could be available 24/7. Umm. And when students ask, uh, this AI different questions related to cybersecurity, uh, they can receive immediate answers that could be tailored to them, whether it be, uh, using certain analogies that. Makes sense to, uh, the user, or if it's. Adapted to. Teach with different styles. Maybe some uh, users prefer a less words, uh, maybe some users prefer more words. And the explanation, uh, how detailed an explanation is. And would this type of personalized learning, uh, AI can adapt to, uh, you know, different explanations they can identify like the weak areas that a user might be facing, maybe. I'm familiar with a certain topic or area that the AI can isolate and help the user address. They could possibly summarize complex concepts. They can create quizzes, generate different examples, and provide different types of feedback. The AI could possibly also, uh, track the progress of the user. Uh, they can probably work with the user to set goals, uh, obviously act as a source of encouragement, uh, and uh, they can keep track of different learning milestones. Back to the certification. The Comtia Security Plus why it matters. It is one of the, uh, world's most widely recognized entry level cybersecurity certifications. So, umm. It's often required for government and defense related cybersecurity positions and is frequently listed in job descriptions. This test, uh, shows uh. Someone's. Ability on network security, uh, shows their skills in threat management, risk management, umm. Security operations. It's basically the test serves as like an industry recognized credential that validates, uh, foundational cybersecurity knowledge. Umm, and if veterans can obtain the certification, umm, they basically demonstrate to employers that indicate. That basically indicates uh. That they possess the central cyber security skills and understands. The. Best practices, uh, for the cybersecurity industry? By obtaining this certification, uh, a lot of opportunities, job related opportunities could open up, whether it be security analytics, uh, information systems, uh, specialists, cybersecurity technicians, uh, System Administrator, network administrator, uh, different types of IT work. Umm, government cybersecurity positions, umm, stuff like that. The project basically aims to combine, uh, artificial intelligence with cybersecurity education to create a more effective and accessible learning experience for veterans, uh, that is hopefully at, uh, little to no cost, umm, by reducing the barriers. Uh, to learn cybersecurity and get the certifications. Umm, the platform can help accelerate a career transitioning, uh, for veterans, uh, and put them in a position where they can enter high demand, uh. Fields. And obtain a strong long term employment opportunities. And the end goal is to after this, they go through this AI accelerator is to have them pass the uh, comp TS security plus. Test uh on their first try.

Requirements and Specifications:

This is for the planning document the final stage as part of the tiger team and my teammates and I have discussed this in depth uh we were wondering whether the final deliverable of the entire project is just an AI digital instructor or a learning excelerator but we understood that we will plan on to the core which is the AI digital instructor the AI digital instructor seems to be the goal uh so on a high level I know that this is a learning instructor and my darn planned document with the rest of my teammates is due July 6th we decided that we're going to submit it together as a team and divide it up into chunks and then later on collaborate entirely for the full project but we're all going to have our raw documents for our sections I know that the target audience are veterans who want to transition into cybersecurity careers we need to do a swat analysis and we also need to write goals and milestones and then we at the end we need AI to evaluate it as doctor williams we also need to make sure to put responsibilities of each team We're going to make sure that we put a responsibility section and then agent deeja will be doing part of the task and schedules B will be in charge of product project description and I will be in charge of the requirements and specifications and we will regroup for our project milestones later on um our next meeting will be tomorrow at 6:20 PM I think for my section in the uh requirements and specifications um So on this responsibilities and requirements and specifications requirements and specifications section I'm thinking that it needs to be low to no cost because the whole point is reducing the massive expenses usually tied to these certifications technically it has to be a genetic thing because it can just spit out static info which has to actually reward reorganize repackage contents so that the users don't just memorize answers but rather understand why what's happening The system is designed especially for veterans transitioning into cyber security careers the tool must accelerate A learner from zero knowledge to certification readiness for the comp Tia security plus exam a primary requirement is that the accelerator must be developed and delivered at little to no cost to the end user the system cannot be a static information repository it needs to have agentic behavior meaning it programmatically programmatically repackages reorganizes and rewards content and adapts itself to ensure deep conceptual mastery rather than work note memorization

Tasks and Schedule + Milestones:

What's gonna happen in this document is that we will be covering the work breakdown structure in order to define tasks for the quantum surge project, basically. So ideally what we would be creating with this project is like an AI learning accelerator to help veterans who are trying to. Which career paths? Develop a sense of confidence and their ability to perform well in the tests that they will be required to take in order to get this, get a certification. So the learning accelerator will basically be like AI model that can help them understand and learn about. Cyber security, which is the field that they will be training in. So the phase one of the project would be to just define the scope itself and. You know, establish learning objectives and, you know, conduct background research as well. And all of these tasks will be part of phase one, which would be just defining the project itself. So after we identify the scope of the project as a whole, as in what parts of. The cyber security network, are we supposed to help them specialize in like? I guess we would have to just look up the CompTIA certification and develop. And understanding a holistic understanding of the course as a whole so that we know how to split up like our learning model. After all of that is done, we will begin designing the system itself, and then we will have to design a chat bot or whatever to be the digital AI. Instructor and developed the architecture behind that and then also develop an agentic workflow structure where. They can go in order to get like ohh. Shoot, I don't know. Anyway. Then you wanna design the instructional methods that they will be learning by. I don't know what that means. I guess it could be like learning modules, it could be like narrative videos, it could be like voice to text or like. that poor its own After that. That will. Then we would be building. The actual prototype. And that will consist of developing the AI. Prompts and then building the workflow and then constructing the modules that they will be learning from. And then we would be developing an evaluation tool like a survey or something where they could. You know, establish a confidence level or like a scale, you know, on the module itself so that they know what to review and where they can go from there. After this prototype development, we will be doing the testing and validation for this. Project where we could conduct baseline assessments. For example, you use data and pretend I am the individual taking the test. I would be. You know, looking at my. Code or my module that I've developed so far and. Establishing a level of confidence in the. Data that I have put together I guess, and then we would also collect performance data. So not just me, but all of my team will be testing the model after it's been. Formed to. Check whether or not it can. Satisfied or deliverables after it has been being has after it has been made. After that we would, you know. Use data analysis, you know, we're learning to identify the positives in our structure, identify the negatives, you know, go through and do the SWOT analysis. We would also follow that up with the glass half empty analysis and just find out where our data could go wrong, where our model. Go wrong and how we could improve that and after that we would come to phase six, which is finalizing our deliverable and. Reporting it. Uh, in this phase, we would complete the final report, we would create the presentation, we would log our progress, and we would compile all of the supporting documentation. involved in this project and in the In the end. Um, and that'll be it. So I'm gonna run through this really quick and define milestones and that could include milestone one, which would be the project scope gets approved. We understand the background of the project, Austin. Two could be the research and all of the intrinsics of the project. The simple function all of that will be finalized I'll Steam 3 would be the system architecture is defined as in we know what code we're going to use what models we're going to use what AI is going to help us build this all of that architecture that goes into building this will be defined milestone Ford will be that the. AI digital instructor prototype itself is completed as in fully functioning, ready to take the load of. Data that it is intended to support and then also if I could do the assessment framework completed as in B, now integrate the questions from the CompTIA certification into the model so that we can perform testing on whether or not it is effective. Houston 6 would be testing. Completed as in the model is fully functioning and it supports the intended usage. Milestone seven could be the data analysis. Completed as in we have collected all of the data required to power this model and that you know it is usable for the veterans you know. Yeah, it could be that the final report is completed. I know we have to create like a textbook, we have to create like a log. All of that would be finished by then. And then milestone 9 would be that the final presentation is delivered where we can show Doctor Williams and other officials what we have developed. She worked. And how it functions and making better impact.